

STROOP EFFECT
PSY310: Lab in Psychology
Lab Report

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GITHUB LINK:

Introduction

Stroop task is a seminal paradigm in Experimental Psychology (Viviani et al, 2023). Stroop task measures the interference between an irrelevant word with a relevant colour. It indicates attentional focus. Studies in the past have reported that patients with psychotic disorders exhibit increased interference (Henik & Salo, 2004)

In a Stroop task the interference is measured by calculating the difference in Response Times (RT) between trials where the font colour and text matched (congruent) versus when it did not match (incongruent). This RT difference is referred to as Stroop effect and is a measure of attentional focus.

Method

Participants: Four students of Ahmedabad University (aged 19–20 years) took part in the experiment. The sample included two females and two males. All participants reported normal or corrected-to-normal vision. Participation was voluntary and anonymous.

Materials and Stimuli: Experiment was performed on a Windows laptop run using PsychoPy. (cite). The stimulus was presented on a 13 inch screen with a refresh rate of 60 Hz. Responses were collected using the standard keyboard. Stimuli consisted of the colour words RED and BLUE, presented in red or blue font colours. Each word subtended a visual angle of approximately 4.77°.

Image 1: Example of an Incongruent Stroop Stimulus

The word “RED” is displayed in blue font colour. Here, the semantic meaning (word) and perceptual feature (font colour) are in conflict, creating interference.

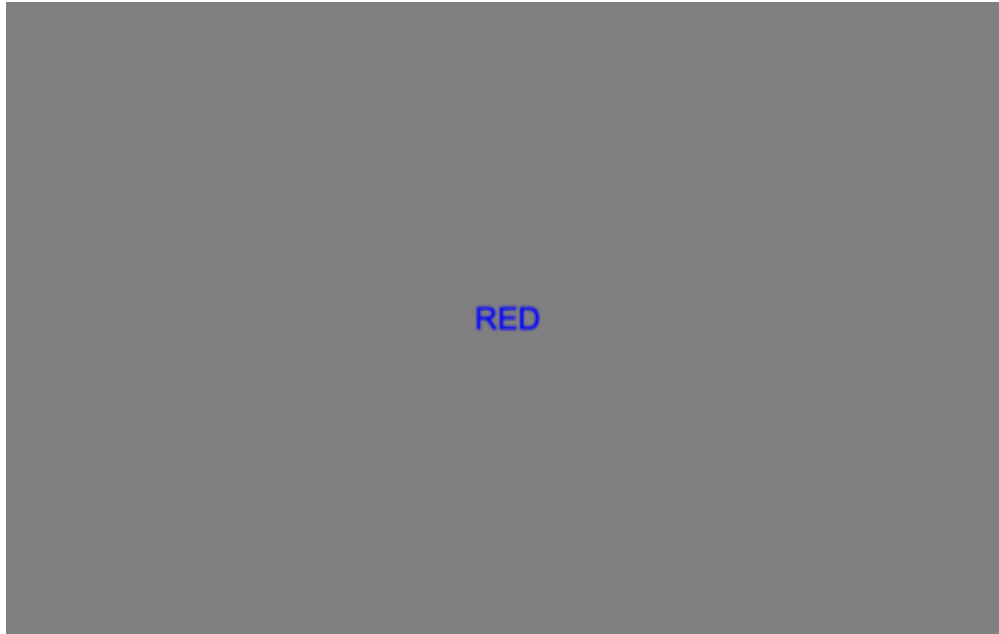
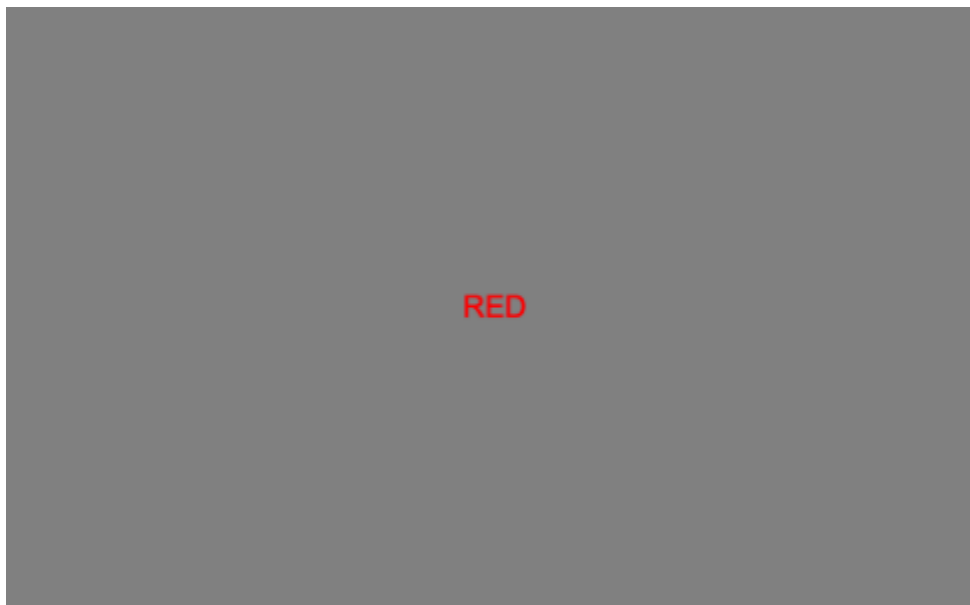


Image 2: Example of a Congruent Stroop Stimulus

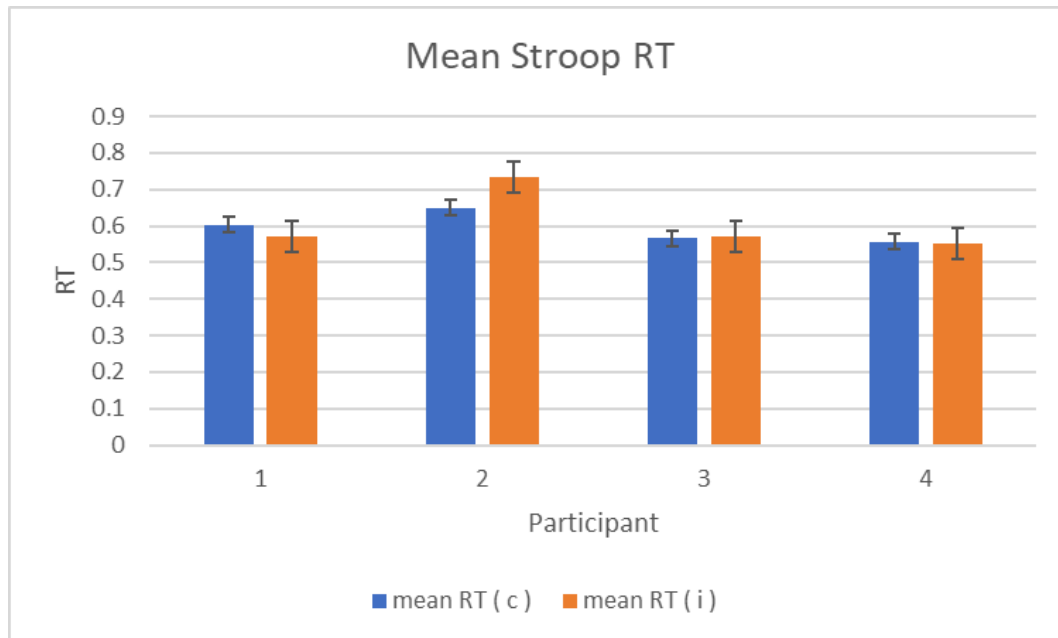
The word “RED” is displayed in red font colour. In this case, the semantic meaning and font colour match, producing faster responses.



Results

Mean correct RTs were put to a paired sample t-test. The test revealed a significant difference between the congruent and incongruent RTs . Congruent RTs (mean=0.594178; SD=0.036938108) were faster than Incongruent RTs (mean= 0.60644075; SD=0.073479705).

The data from the experiment is summarized in the graph below.



Discussion

The results show that, on average, participants were slower in incongruent trials than in congruent trials. This pattern is consistent with the Stroop effect, which suggests that conflicting semantic information increases the difficulty of naming colours.

However, the Stroop effect was small (approximately 19 ms) and not statistically significant. This is likely due to the small sample size ($n=3$), which reduces statistical power and makes it difficult to detect reliable differences. Despite this, the direction of the results aligns with the classic Stroop effect. These findings are consistent with past research that identifies Stroop interference as a measure of attentional focus.

Limitations

Limitations of this study include the restricted sample size, which severely limits generalizability, and the use of only two colours and two words, which may reduce the robustness of the effect compared to full Stroop paradigms with multiple colour-word combinations.

References

- Henik, A., & Salo, R. (2004). Schizophrenia and the Stroop Effect. *Behavioral and Cognitive Neuroscience Reviews*, 3(1), 42-59.
<https://doi.org/10.1177/1534582304263252>
- Viviani, R., Spahn, A., Neuschwander, M., & Beschoner, P. (2023). Stroop task and measures of interference: A meta-analysis. *Psychological Research*, 87(3), 1230-1245.